

I/WE CLAIM

INDEPENDENT CLAIMS

Claim 1: A self-evolving adaptive cognitive data fabric system comprising:

[0117] a quantum-resistant data collector configured for secure, multi-protocol ingestion with dynamic key rotation and post-quantum cryptographic algorithms;

[0118] a stream reconciliation engine that implements self-healing consistency with Byzantine fault tolerance to reliably synchronize heterogeneous data streams;

[0119] a real-time semantic profiler that utilizes differential privacy-preserving vector embedding generation by integrating natural language processing and computer vision for multi-modal data analysis;

[0120] an adaptive pipeline orchestrator employing a multi-agent reinforcement learning framework with a hierarchical reward system and automatic feature engineering based on semantic feedback;

[0121] a cognitive model re-trainer configured for incremental knowledge distillation with continual learning and curriculum-based retraining to address concept drift; and

[0122] a federated metadata manager that enforces immutable, blockchain-based provenance tracking with zero-knowledge proofs.

Claim 2: A method for dynamic optimization of data processing workflows in an adaptive cognitive data fabric, comprising:

[0123] continuously ingesting and reconciling heterogeneous data streams via a quantum-resistant data collector and self-healing reconciliation engine;

[0124] analyzing the ingested data through a differential privacy-preserving semantic profiler to extract schema, content, and multi-modal semantic meaning;

[0125] dynamically reconfiguring processing workflows using an adaptive pipeline orchestrator that leverages multi-agent reinforcement learning with a hierarchical reward mechanism and automatic feature engineering;

[0126] triggering retraining of cognitive models through incremental knowledge distillation and curriculum-based prioritization upon detecting performance drift; and

[0127] maintaining immutable metadata and provenance records via a federated governance framework employing blockchain-based verification and zero-knowledge proofs.

Claim 3: A self-adaptive federated governance and policy enforcement system for a cognitive data fabric, comprising:

[0128] a federated metadata manager that maintains a centralized yet distributed repository of data catalogs with blockchain-based immutable tracking and zero-knowledge proofs;

[0129] a policy enforcement engine that translates natural language policies into formally verified, rule-based implementations and utilizes homomorphic encryption for real-time access control;

[0130] an explainability service that generates counterfactual explanations through causal model extraction to enhance decision transparency; and

[0131] a resource optimization module integrating a market-based predictive resource allocator, dynamic scaling engine, topology-aware peer discovery for secure P2P mesh communication, and carbon-aware scheduling to optimize computing resources across heterogeneous environments.

DEPENDENT CLAIMS:

[0132] The system of Final Independent Claim 1, wherein the adaptive pipeline orchestrator employs a multi-agent reinforcement learning framework with a hierarchical reward system balancing immediate processing efficiency, computational constraints, and long-term learning objectives.

[0133] The system of Final Independent Claim 2, wherein the real-time semantic profiler implements multi-modal vector embedding alignment that dynamically adjusts embedding dimensions based on semantic similarity metrics to maintain a unified representation across diverse data types.

[0134] The system of Final Independent Claim 1, wherein the cognitive model re-trainer implements curriculum learning to prioritize retraining on samples that most effectively

address performance gaps while maintaining a directed acyclic graph (DAG) of model derivations for automated performance comparison.

[0135] The system of Final Independent Claim 3, wherein the explainability service generates counterfactual explanations using an invertible transformation map between input features and model decisions to enable bidirectional traceability of decision pathways.

[0136] The system of Final Independent Claim 1, wherein the federated metadata manager implements cross-jurisdictional compliance verification using selective disclosure protocols integrated with blockchain-based immutable record-keeping.

[0137] The system of Final Independent Claim 3, wherein the topology-aware peer discovery system implements a self-organizing compute mesh that dynamically redistributes workloads based on network conditions, hardware capabilities, and energy efficiency metrics.

[0138] The system of Final Independent Claim 1, wherein the quantum-resistant data collector incorporates forward secrecy protocols with post-quantum cryptographic primitives to ensure robust security against future quantum computing attacks.

[0139] The system of Final Independent Claim 3, wherein the natural language policy compiler automatically translates high-level governance policies into formally verified, rule-based enforcement mechanisms with mathematical guarantees of compliance across diverse technology stacks.